EVERY FINITELY BRANCHING LEVEL TREE CARRIES A SEQUENCE OF ORDERED PARTITIONS WITH UNIFORM SUCCESSOR NUMBERS

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Editorial note. This is the sixth card this lane authors from this source and the last remaining low-cost candidate there; it was taken to keep the publishable queue fed before moving on to longer proofs.

Allowed standard prerequisites (editorial).

- The levels L_k of a tree with omega levels, all finite, and the successor set of a node. Set up in the paragraphs preceding the counted statement

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(ending at source line 525). Fixes the objects that are partitioned and the successor counts that have to become uniform.

- A sequence of natural numbers in which every number occurs infinitely often, and the eventual constancy of a refining sequence of partitions of a finite set. Used inside the counted proof at lines 532 and 546-547. Supplies the schedule of refinements and the reason the refining sequences stabilise.

Theorem 0.1. *Let $(T, \leq)$ be a tree with ω levels, all finite, with kth level L_k. Then there is a sequence of ordered partitions $((\pi_k, <) : k \in \omega)$ such that π_k is a partition of L_k, and such that for any $X \in \pi_k$ and $Y \in \pi_{k+1}$ there is $n(X, Y) \in \omega$ such that for each $x \in X$ the number of successors of x lying in Y is equal to $n(X, Y)$.*

Proof. We use the same sequence (k_n) as in the previous proof, in which each member of ω appears infinitely often, and for each k we find a refining sequence of partitions $((\pi_k^n, <) : n \in \omega)$ of L_k, which here is taken to mean that each member of π_k^n is a union of a convex subset of π_k^{n+1}. We start with each π_k^0 being the trivial partition of L_k into just one piece.

Now assume that $(\pi_k^n, <)$ has been chosen for each k. If $k \neq k_n$, then $\pi_k^{n+1} = \pi_k^n$. So we just have to define π_k^{n+1} (and a suitable ordering of it) in the case that $k = k_n$. For each $x \in L_k$, and $X \in \pi_{k+1}^n$, consider the sequence of natural numbers $(|X \cap succ(x)| : X \in \pi_{k+1}^n)$ of length the cardinality of π_{k+1}^n and let $x \sim y$ if the sequences corresponding to x and y are equal. We then take π_k^{n+1} to be the least common refinement of π_k^n and the partition determined by $\sim$. We can (definably) linearly order the finite sequences of natural numbers and hence extend the ordering of π_k^n to $\pi_k^{n+1} \restriction X$ for each $X \in \pi_k^n$ which will thus form a convex subset.

Since $(\pi_k^n : n \in \omega)$ is a refining sequence and L_k is finite, it must be eventually constant, at π_k say. The relationship between π_k and π_{k+1} may now be described. Clearly as the sequence has stabilized, $\sim$ must be trivial on each member of π_k. So if $X \in \pi_k$, $(|Y \cap succ(x)| : Y \in \pi_{k+1})$ is the same sequence for each $x \in X$. Since $|succ(x)|$ is the sum of this sequence, this means in particular that each $x \in X$ has the same number of successors. And for each $Y \in \pi_{k+1}$, each $x \in X$ has the same number of successors in Y, which we write as $n(X, Y)$ (which is allowed to be 0). □